TECHNICAL PRODUCT REQUIREMENTS SPEC

AI Founder OS — Sales Copilot

Target Audience: AI Developers & Interns
Domain: www.bluecode-os.com
Version 1.0 (MVP)

1 EXECUTIVE OVERVIEW & PLATFORM CORE

AI Founder OS is a multi-tenant, web-based SaaS platform empowering early-stage startups with an autonomous AI-powered sales team. The core initial offering—the **AI Sales Copilot**—automates customer acquisition from Ideal Customer Profile (ICP) definition to meeting booking.

Multi-Tenant SaaS

Strict data isolation per startup account. Startups log in to dedicated workspaces to manage their unique ICPs, leads, and outreach campaigns.

Hosting & Infrastructure

Phase 1: Hostinger VPS (Shared Node setup) for cost-efficiency.
Phase 2: AWS Cloud migration post-investment funding.

Billing & Payments

Integrated **Stripe** gateway handling domestic (INR ₹) and international (USD $) subscriptions, tracking tier limits and usage.

2 END-TO-END SALES PIPELINE WORKFLOW

The AI Sales Copilot drives prospects seamlessly through an autonomous 8-stage pipeline:

1. ICP CONFIG Target parameters → **2. DISCOVERY** Company search → **3. DM IDENTIFICATION** Exec lookup → **4. RESEARCH & SCORE** AI Scoring (0-100) → **5. OUTREACH** Personalized emails → **6. NUDGE** Follow-up bot → **7. BOOKING** Calendar Sync

3 PLATFORM CAPABILITY MATRIX

Module Name	Core Capability Description	MVP Inclusion
1. ICP Builder	Configures target profile (Industry, company size, geography, job titles, exclusions).	MVP Phase 1
2. Prospect Finder	Queries databases to surface companies matching configured ICP parameters.	MVP Phase 1
3. DM Finder	Identifies verified key decision-makers (Founders, CEOs, HR Heads, CTOs).	MVP Phase 1
4. Company Research	Scrapes and summarizes target firm's offerings, recent news, website, and tech stack.	Post-MVP Phase
5. Lead Scoring	Algorithms evaluate match relevance and assign 0-100 score + text rationale.	MVP Phase 1
6. Outreach Engine	Generates customized cold emails and multi-touch Apollo messaging sequences.	MVP Phase 1
7. Follow-up Assistant	Tracks response inactivity and generates context-aware follow-up prompts.	MVP Phase 1
8. Meeting Scheduler	Integrates calendar booking links automatically into emails/messages upon positive replies.	MVP Phase 1
9. Prioritization Engine	Ranks daily sales actions for founders based on engagement & lead priority.	Post-MVP Phase
10. Sales Dashboard	Centralized metric overview of pipeline stages, response rates, and	MVP Phase 1

4 MVP SCOPE: DETAILED FUNCTIONAL SPECIFICATIONS

To launch rapidly without over-engineering, the MVP is strictly scoped to **7 core functions**. Below are the functional requirements for each module:

1. ICP Builder Module

Allows founders to enter natural language descriptions of their product. The AI engine parses input and structured controls to construct:

- Target Industry & Vertical
- Company Headcount (e.g., 50–500 employees)
- Target Geographies & Language
- Decision-Maker Job Titles (e.g., HR Head, CEO)
- Negative Exclusion Criteria

2. Prospect & DM Finder

Retrieves target companies and verified contact profiles:

- Queries external data APIs (Apollo / Third-party data brokers).
- Extracts full name, verified business email, LinkedIn URL, and verified job title.
- Associates identified contacts directly with their company record.

3. AI Lead Scoring Engine

Evaluates lead match quality and assigns a numerical score with explicit AI-generated reasoning:

- `90–100 Hot` Perfect ICP fit & clear triggers.
- `70–89 High` Strong match across key criteria.
- `50–69 Medium` Partial match; secondary priority.
- `< 50 Low` Poor fit; auto-filtered out.

4. Personalized Outreach Generator

Generates high-converting, personalized cold sequences based on company and decision-maker profile:

- Personalized Cold Email (Subject line + body).
- Apollo Direct Message sequence.
- Follow-up #1 (Bump / Value add).
- Follow-up #2 (Break-up / Final call).

5. Follow-up Assistant

Monitors inbox activity and alerts founders on required touchpoints:

- Tracks message dispatch timestamps.
- Triggers automated alerts: *"18 prospects require follow-up today."*
- Pre-generates contextual reply copy.

6. Auto-Scheduling Integration

Converts positive replies into booked sales meetings:

- Integrates Email SMTP/IMAP & Apollo APIs.
- Detects positive intent in incoming responses.
- Appends startup calendar links (Calendly/Custom) to book meetings automatically.

7. Centralized Sales Dashboard (UI/UX Mockup Flow)

The single-page UI view must summarize daily actionable metrics for the logged-in startup:

```
[ Dashboard View — Welcome, ABC Technologies ]
ICP: Indian B2B Companies | 50-500 Employees | Target: HR Heads / Founders
--------------------------------------------------------------------------------
TODAY'S ACTIONS: 🔎 25 Prospects Found | 🎯 12 Hot Leads | 📧 10 Messages Ready | 🔄 8 Follow-ups | 📅 3 Meetings Booked
```

5 SYSTEM ARCHITECTURE & TECH STACK

Tech Stack Specifications

- **Frontend:** React / Next.js (Tailwind CSS, clean dashboard UI).
- **Backend:** Node.js (Express) or Python (FastAPI) microservices.
- **AI Orchestration:** OpenAI API / Anthropic Claude API via LangChain or LlamaIndex.
- **Database:** PostgreSQL with Row-Level Security (RLS) or MongoDB multi-tenant collections.
- **Integrations:** Apollo.io API, SendGrid / SMTP, Stripe SDK.

Frontend Navigation Blueprint

```
AI SALES COPILOT (UI Layout)
├── 📊 Dashboard (Action center)
├── 🎯 ICP Builder (Profile setup)
├── 🔍 Prospects (Search & Research)
├── 🏆 Leads (Scored & Ranked)
├── 👥 Decision Makers (Contact list)
├── ✉️ Outreach (Email & Apollo)
├── 🔄 Follow-ups (Pending queues)
├── 📈 Pipeline & Analytics
└── ⚙️ Settings (Company, Users, Billing)
```

6 MULTI-TENANT DATABASE ARCHITECTURE

Data isolation is enforced at the database level. Every query must filter by `tenant_id` (`startup_id`) to ensure zero cross-tenant data leaks.

Data Model Hierarchy

```
AI_Sales_Copilot_DB
│
├── Startup Accounts (tenant_id, plan, status)
│   ├── Users (user_id, role, credentials)
│   ├── ICPs (icp_id, criteria_json)
│   ├── Prospects (prospect_id, firmographics)
│   │   └── Decision_Makers (dm_id, email, role)
│   ├── Leads (lead_id, score, rationale)
│   ├── Outreach_History (msg_id, status, logs)
│   └── Usage_Tracker (api_spend, limits)
```

Tenant Isolation & Security Rules

- **Isolation:** Each startup owns isolated records for Users, Prospects, Leads, and Messaging history.
- **Auth Middleware:** Validate JWT bearer tokens on all API routes; inject `startup_id` context into DB queries.
- **API Rate Guardrails:** Middleware checks `Usage_Tracker` before invoking LLM or Apollo APIs to prevent budget overruns.

Backend Intern Instruction: Always implement strict validation on foreign keys (`startup_id`). Write unit tests verifying that User from `Startup A` cannot access API endpoints returning data for `Startup B`.

7 SUBSCRIPTION TIERS & COST GUARDRAILS

To ensure high profitability, third-party AI/API token costs are capped per subscription tier. Software pricing yields a guaranteed **60% gross profit margin** across all paid tiers.

Subscription Package	Pricing (INR / USD)	API Spend Cap (Max Cost)	Net Profit / Margin	Key Tier Entitlements
15-Day Free Trial *(First 20 Startups)*	**Free** ₹0 / $0	₹1,000 max *(Hard cap limit)*	- ₹1,000 *(Customer Acq. Cost)*	1 User | 1 ICP | 50 Research Credits | Basic Outreach | 15-day limit
Starter Tier	**₹2,500 / mo** ($29 / mo)	₹1,000 / mo	**₹1,500 profit** (60% Margin)	1 User | 1 ICP | 100 AI Prospects | 50 DM Searches | 100 Emails
Growth Tier	**₹5,000 / mo** ($59 / mo)	₹2,000 / mo	**₹3,000 profit** (60% Margin)	3 Users | 3 ICPs | 500 AI Prospects | 250 DM Searches | CRM Sync
Pro Tier	**₹10,000 / mo** ($99 / mo)	₹5,000 / mo	**₹5,000 profit** (50% Margin)	10 Users | 10 ICPs | 2000 AI Prospects | 1000 DMs | Priority Support

8 PHASED DEVELOPER EXECUTION ROADMAP

Phase 1: Pilot MVP (Immediate Sprint)

Objective: Onboard first 15–20 startups on a single common plan with a 15-day free trial to gather real usage data.

- Build core 7 MVP modules.
- Deploy single common subscription tier logic.
- Implement hard API spend cap (₹1,000 / startup).
- Track actual usage metrics: *Do founders use 50 vs 500 prospects? Do they send emails vs LinkedIn messages?*

Phase 2: Scale & Tier Optimization

Objective: Deploy full multi-tier SaaS platform based on empirical pilot data.

- Activate Stripe automated tier upgrades (Starter/Growth/Pro).
- Build advanced analytics and CRM export capabilities.
- Migrate hosting infrastructure from Hostinger VPS to AWS.
- Introduce usage-based add-on credit packs.

Summary Goal for AI Interns: Deliver Phase 1 MVP code with robust error handling for API call failures, strict data isolation per startup account, and clean user-facing status indicators during AI prospect generation.